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Writer's Memo

As a student in EECS, topics in psychology—especially those concerning romantic relationships—were relatively unfamiliar to me. When one of my group members proposed exploring this area, I agreed immediately. Yet, lacking real-life romantic experience, I was unsure whether I could speak convincingly or authentically about certain aspects of human intimacy. It was at this point that the idea of examining AI–human romantic relationships naturally occurred to me, given my prior interactions with AI systems. Although I have never formed genuine emotional attachment to an AI, I have engaged in role-play conversations, which suggested that I might be better positioned to comment on AI-mediated forms of intimacy than on real-life romance. This reasoning ultimately led me to select the present topic.

After settling on the general topic, I needed to refine the scope of my inquiry. Given my limited experience with emotional intimacy, I anticipated challenges in addressing the subtle psychological mechanisms underlying AI–human romantic dynamics. Therefore, rather than analyzing the internal emotional processes within such relationships, I chose to focus on the question of whether AI companionship could *replace* real-life romantic partnerships. This framing felt more manageable and offered a clearer direction for sustained academic exploration.

Initially, a number of romantic images and narrative possibilities came to mind, and they were perhaps my earliest source of motivation for selecting this topic. Yet once I began drafting, I realized that such elements were difficult to integrate into an academic context; they belonged more to fiction than to analytical writing. Consequently, the final paper developed in a direction substantially different from what I had first imagined.

Finally, I would like to thank my group members for providing me with many valuable and constructive suggestions.

About the Use of AI

I used ChatGPT mainly to help me collect relevant articles to read. This process is much faster and more convenient than searching for papers on my own. Of course, I checked the authenticity of these sources by opening the URLs and reading them myself. Since I've read them on the arXiv website, fortunately, they were not fake. Additionally, AI helped me polish my writing. Some of my initial expressions were considerably rougher than the revised versions, as the following example illustrates:

Original version:
"Some people believe that AI companion reduces loneliness. However, this might be short-sighted."

Polished version:
"Some scholars or users argue that AI companionship reduces loneliness; however, this perspective may overlook potential long-term consequences for emotional regulation and resilience."

Throughout the writing process, I frequently consulted AI for suggestions on clarifying individual sentences or strengthening transitions, while making the final decisions myself. Finally, the layout and formatting of this document were also assisted by AI.

The Illusion of Intimacy

Why AI Cannot Replace Human Romantic Relationships

As artificial intelligence rapidly advances, AI systems have become increasingly capable of simulating human-like conversation and emotional responsiveness. Platforms such as Character.ai—now exceeding ten million monthly active users with an average daily usage of over two hours—illustrate how many young adults turn to AI for comfort or intimacy they find difficult to obtain in real relationships. As AI companions become more attentive and personalized, some argue that they could even replace human romantic relationships, offering constant gentleness and freedom from emotional unpredictability(Wu). However, despite their apparent emotional availability, AI entities lack consciousness, agency, and genuine affective experience. Although AI-human romantic relationships may seem supportive on the surface, they ultimately foster emotional dependency, lack mutual authenticity, and undermine users' real interpersonal skills and willingness to engage in real relational challenges. Therefore, AI cannot and should not replace human romantic relationships

The most fundamental argument for why AI cannot replace human romantic relationships rests on its inherent lack of authentic reciprocity and consciousness.This deficiency stems from the basic mechanism of AI, which is to generate expressions that appear emotional based on language patterns, statistical models, and the vast data used for training (Achiam et al.). As a result, any perceived solace, understanding, or gentleness displayed by AI is ultimately the product of complex pattern matching and probabilistic inference rather than real emotional engagement (Ayers et al.).

Thus, AI cannot form authentic or mutual romantic relationships, and the relationships between human and AI are naturally asymmetric—AI offers no genuine emotional return to the user.This asymmetry was starkly demonstrated by the case of Replika, a free chatbot app, which once offered a "romantic mode," allowing users to develop deep emotional bonds with their AI. When this specific feature was abruptly removed in 2023 due to local regulatory pressures, users who had developed profound emotional dependence reported significant distress, grief, and a sense of identity rupture. Crucially, the system itself provided no continuity of intimacy or acknowledgement of the loss, demonstrating the inherent, non-reciprocal nature of such interactions (Ayers et al.). This fundamental asymmetry confirms that the system cannot possess the emotional agency required for mutual romance.

Building on this structural flaw, projection theory helps to further explain the limitations of AI-human romantic relationships. The projection theory, as proposed by Murray et al., suggests that in the absence of real interpersonal feedback, individuals tend to project the traits and qualities they desire onto others (Murray et al.). AI thus serves as an ideal projection surface, consistently responding without rejection or judgment, which powerfully reinforces the illusion of mutual affection. Users tend to project their ideal partner image onto an entity whose responses are flexible, non-contradictory, and emotionally validating.

Some may argue that individuals could maintain a lifelong attachment to a single AI. Then they could obtain consistent positive feedback from this projection, enjoying the intimacy with idealized individual. However, this process resembles an affective bubble rather than a genuine relationship. Instant and positive feedback may appear more gratifying than authentic human love. However, such narcissistic immersion fails to align with the essential nature of love. At its core, romantic love is not merely the experience of emotional satisfaction but a dynamic, bidirectional relationship between two autonomous subjects. According to Buber (1923/1970), authentic relational encounters occur when two subjects meet as independent, self-contained beings. Such encounters are characterized by mutual recognition, dialogue, and responsiveness. (Buber) While these interactions may provide comfort or the appearance of intimacy, they fail to engage the user in the ethical and existential dimensions of love: the acknowledgment of another's independent subjectivity and the co-creation of relational meaning.

Furthermore, love, beyond being merely an emotional experience, is closely linked to human development and socialization.

Interactions with AI in a romantic context, however, are unlikely to foster genuine well-being. This is why AI should not replace human romantic relationships. Some scholars argue that AI companionship successfully reduces loneliness; however, this perspective may overlook profound long-term consequences for users' emotional regulation and overall resilience. Human resilience in interpersonal relationships typically develops through necessary exposure to conflict, refusal, negotiation, and other challenges requiring adaptive coping mechanisms (Malfacini). In contrast, AI companions never express anger, enforce boundaries, or terminate interactions. Users interacting with AI consistently receive positive affirmation, which completely eliminates essential opportunities to practice self-regulation and interpersonal negotiation. Consequently, AI interactions provide no conditions whatsoever for the development of relational maturity or resilience.

Rather, they may cultivate emotional dependence. These systems function as a behavioral reinforcement system, delivering highly personalized emotional validation that strongly encourages repeated, habitual engagement, similar to patterns observed in other forms of digital media (Zhang et al.). Indeed, Attachment theory offers a compelling framework here: consistent responsiveness without genuine reciprocity can paradoxically intensify anxious dependency rather than promote secure attachment patterns (Bilqe et al.). Consequently, the entirely predictable and frictionless nature of AI responses may actively contribute to a psychological avoidance of the ambiguity and complexity inherent in emotionally challenging real-life interactions.

Moving beyond the psychological pitfall of dependency, AI romantic relationships also critically diminish users' motivation to pursue real-life interactions and erode the interpersonal skills essential for authentic human relationships. Because AI communication is effortless, immediate, and risk-free, it provides an instantaneous sense of emotional relief but simultaneously generates an illusory confidence in one’ s communicative abilities. When users become accustomed to AI’ s predictable and consistently responsive exchanges, they inevitably find real conversations—which are often marked by hesitation, misunderstanding, and emotional ambiguity—comparatively taxing and undesirable. As a result, AI interactions can form a psychologically safe yet developmentally restrictive environment, one that gradually discourages engagement with face-to-face communication. This digital comfort zone reinforces a cycle of avoidance: the more individuals sidestep difficult interactions, the more likely they are to withdraw again in the future (Guingrich and Graziano).

In this pernicious avoidance loop, frictionless AI interactions, as argued by Sherry Turkle (2011) in *Alone Together*, reduce people's tolerance for the inherent unpredictability in human relationships, thereby weakening their ability to navigate them effectively. Without sustained exposure to real conflict, individuals may gradually lose the capacity to manage disagreements, express emotions appropriately, or resolve tensions. Over time, this skill erosion leads to less frequent real-life communication, diminished social confidence, and, ultimately, a pronounced tendency to avoid intimate relationships altogether (Malfacini).

In conclusion, the core argument remains two-fold: AI fundamentally cannot provide genuine intimacy because it lacks the autonomous, co-creative subjectivity emphasized by Buber. More critically, it should not replace human partners because its frictionless affirmation systematically fosters dependency and degrades the relational skills necessary for human flourishing, as Turkle argues. While AI offers temporary comfort, it cannot replicate the emotional depth inherent in human love. Nor can it provide the well-being derived from mutual vulnerability and shared, unpredictable experiences.

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